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**Research topic: Reddit Users' Reaction to Major US
Political Events in 2020-2021**

Agenda

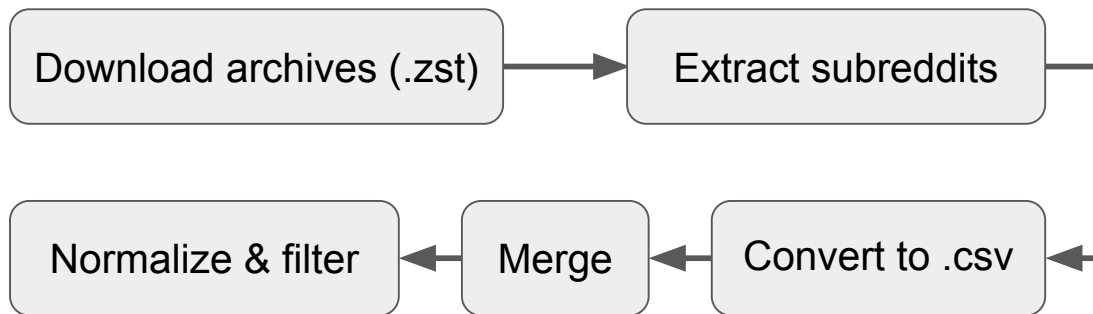
1. Data collection
 - a. process overview
 - b. metrics and challenges
2. Exploratory Data Analysis Workflow
 - a. our approach
 - b. first questions and research developing
 - c. our main visualizations and interpretations
 - d. what required deeper investigation
3. Final Insights
4. Future Work
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Data collection

Source & tools used:

- Data downloaded from Academic Torrents
- Processing was done via Python scripts: `combine_folder_multiprocess.py` for extraction and `parallel_converter.py` to convert from `.zst` to `.csv`

Collection pipeline:



Targeted subreddits:

- Liberal: `r/politics`, `r/democrats`, `r/politicaldiscussion`
- Conservative: `r/conservative`, `r/republicans`, `r/conspiracy`
- Neutral: `r/news`, `r/worldnews`, `r/neutralpolitics`

Timeframe:

- Range: October 2020 - March 2021
- Main focus: the 2020 US Presidential Election and the January 6th Capitol Attack

Dataset metrics & challenges

Unprocessed dataset

Total size (mb)	Total rows	Submissions Count	Comments Count
6886.5	37433549	1139662	26293503

Processed dataset

Total size (mb)	Total rows	Submissions Count	Comments Count
2055.3	5917521	325002	5592519

Challenges and time spent:

- Volume: downloading and handling massive .zst archives (total 120GB) required large amount of disk space and parallel processing to avoid memory overflows
- Data cleaning: we spent significant time filtering dataset from useless content
- Wait times: torrent downloads and script execution took large amount of time due to file sizes
- Time spent: 24 hours totally

Exploratory Data Analysis Workflow

Our approach:

We divided users into liberal, conservative, neutral groups and analysed three key events: 2020 US elections, Capitol raid and Impeachment.

Initial questions:

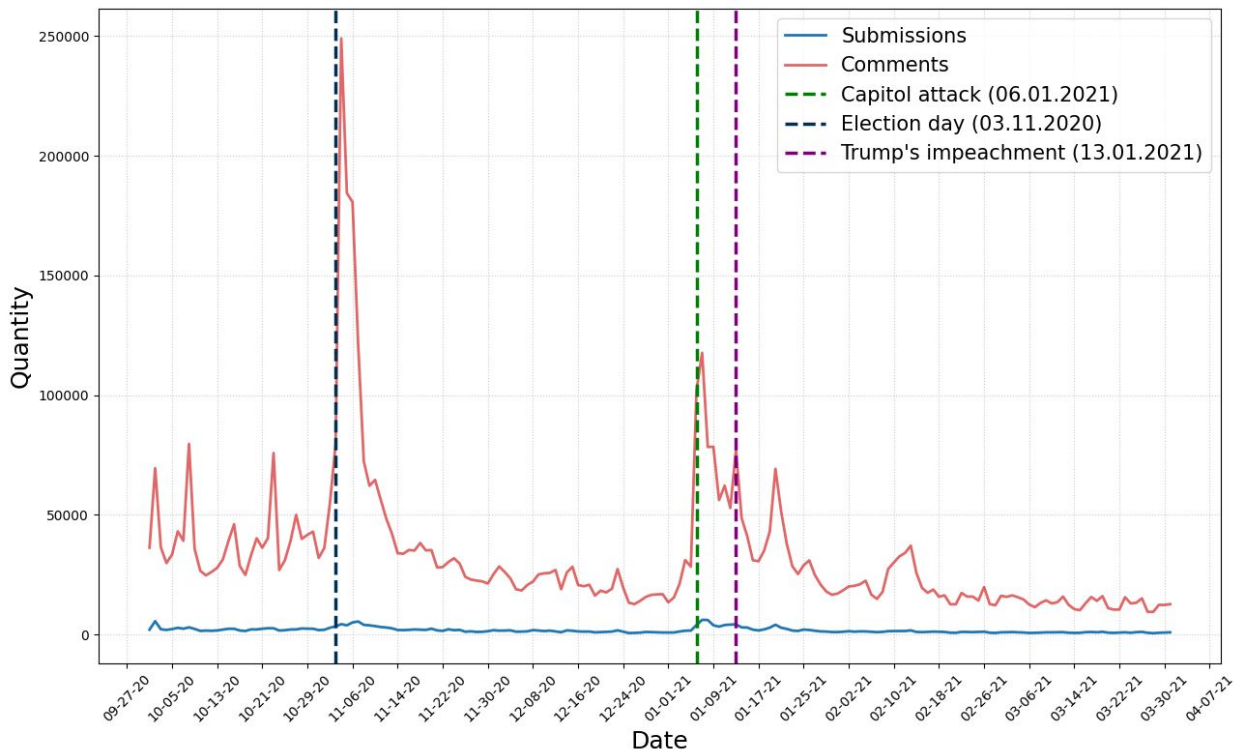
- How activity is distributed inside the political groups?
- Which group generates the highest volume of discourse during peak events?
- What is the sentiment of each group?
- Did sentiment change significantly after the Capitol attack?
- What are the main discussed topics?
- Are comment topics influenced by submission topics?
- Do Liberals and Conservatives speak different language?
- What words defined each group?

How our ideas evolved during EDA

Initial question	Evolved question
How activity is distributed inside political groups?	Temporal distribution and share of voice on critical dates (Election, Capitol Attack)
Was the main topic persistent?	Topic Weight analysis to track the importance of secondary themes (like Covid-19) across the three key events
What is the sentiment of the group?	What is the sentiment of topics inside each group?
Which groups generates the highest volume of discourse?	What is the distance between topics and how topics connected inside a group?

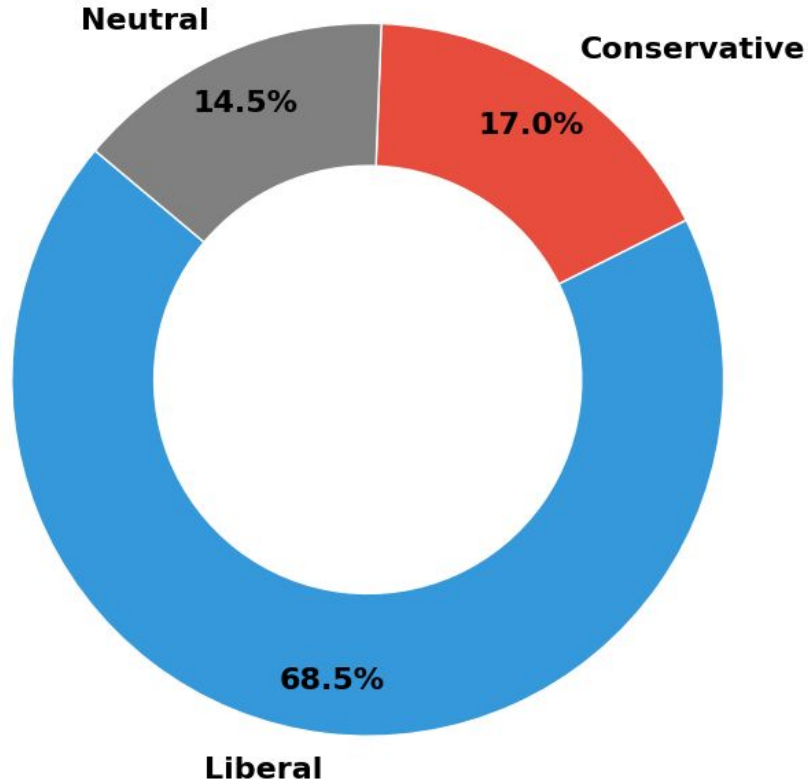
Our main visualizations.

Daily activity volume on key political events



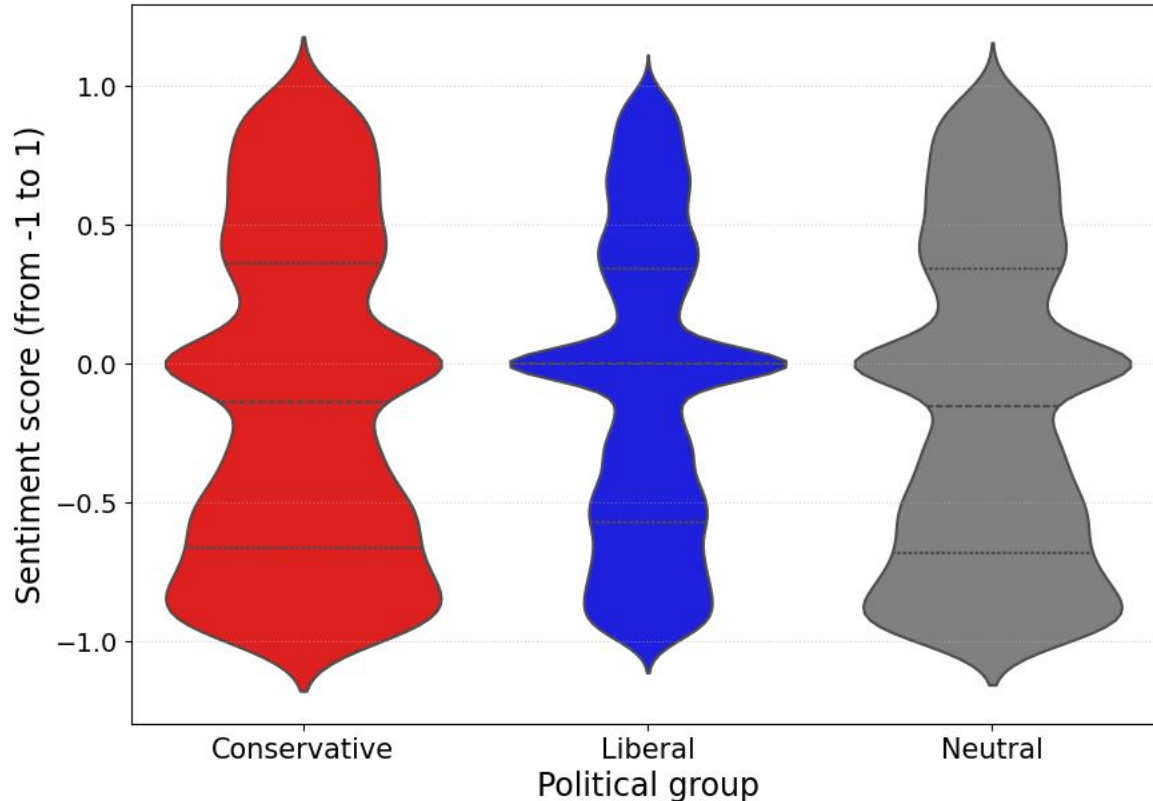
- Event-driven spikes: the two highest peaks align with the Election (Nov 3) and the Capitol Attack (Jan 6)
- The Election spike was wider (persistent discussion), while the Capitol attack spike was sharper and narrower which indicates about massive sudden reaction to the violence

Comment volume by political group



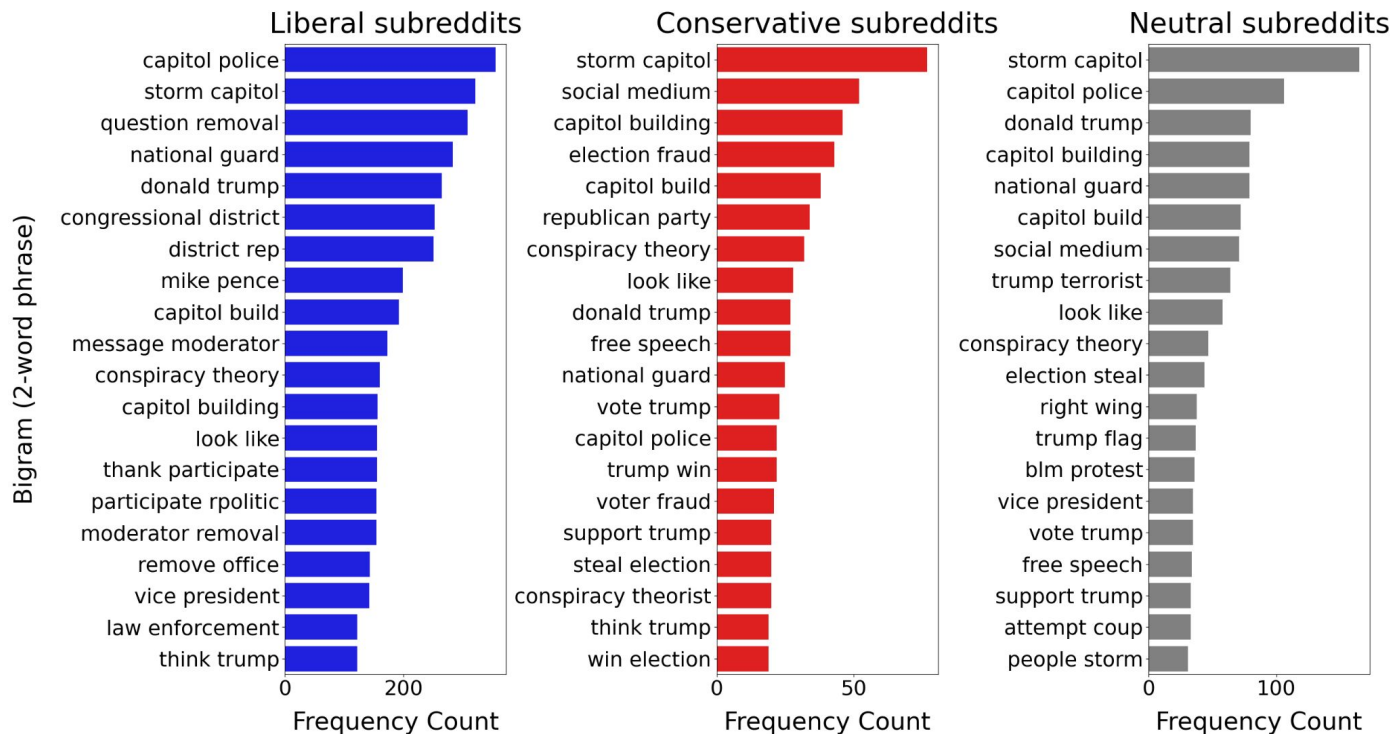
- The dominance of Liberal comments confirms that Reddit isn't a neutral place but systematically unbalanced environment. So any "average" metric calculated for Reddit is mostly Liberal metric
- The Conservative smaller comment volume makes it vulnerable to external noise from Liberals.

Frequency of sentiment distribution by political group (Capitol attack day)



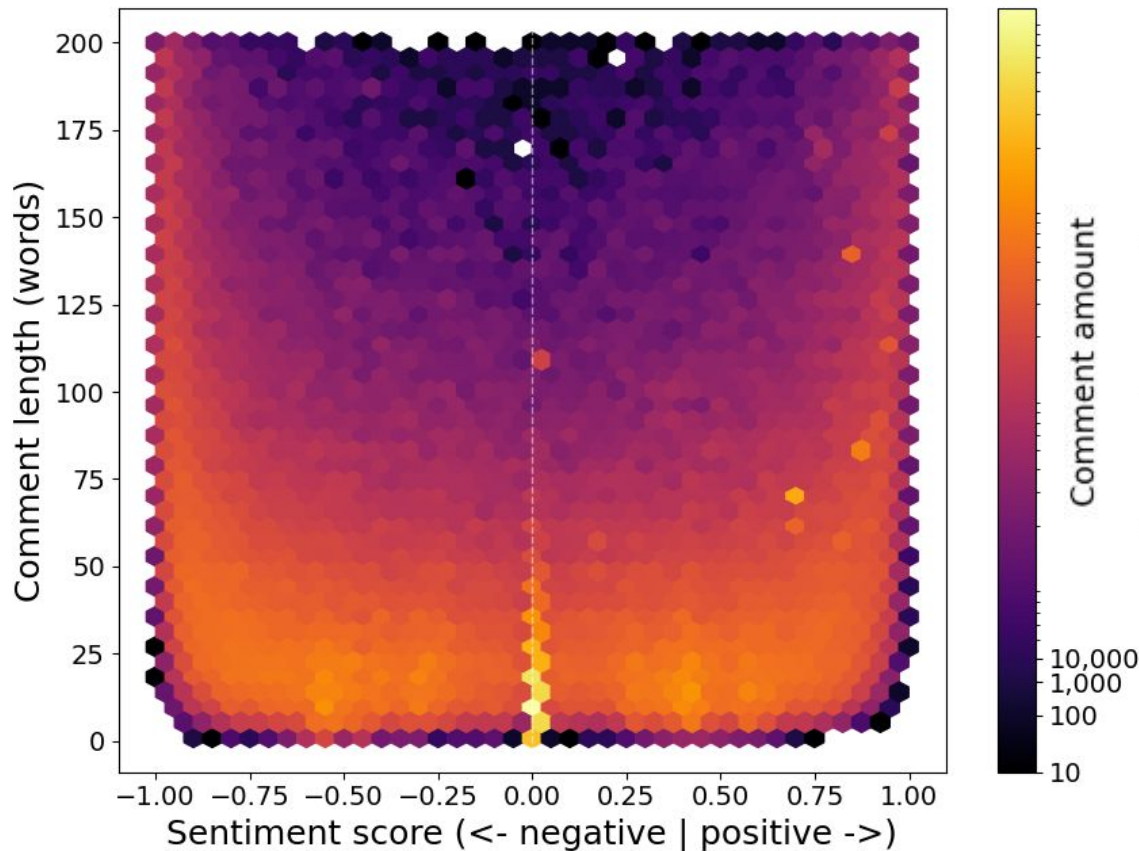
- Though Conservatives make only 17% of comments of the total dataset (compared to 68.5% for Liberals), their sentiment frequency on this day was massive.
- Even the Neutral group reflects this polarization. This reveals that the event was so shocking so even neutral observers started using emotional vocabulary

The most frequent bigrams in comments on Capitol Attack day



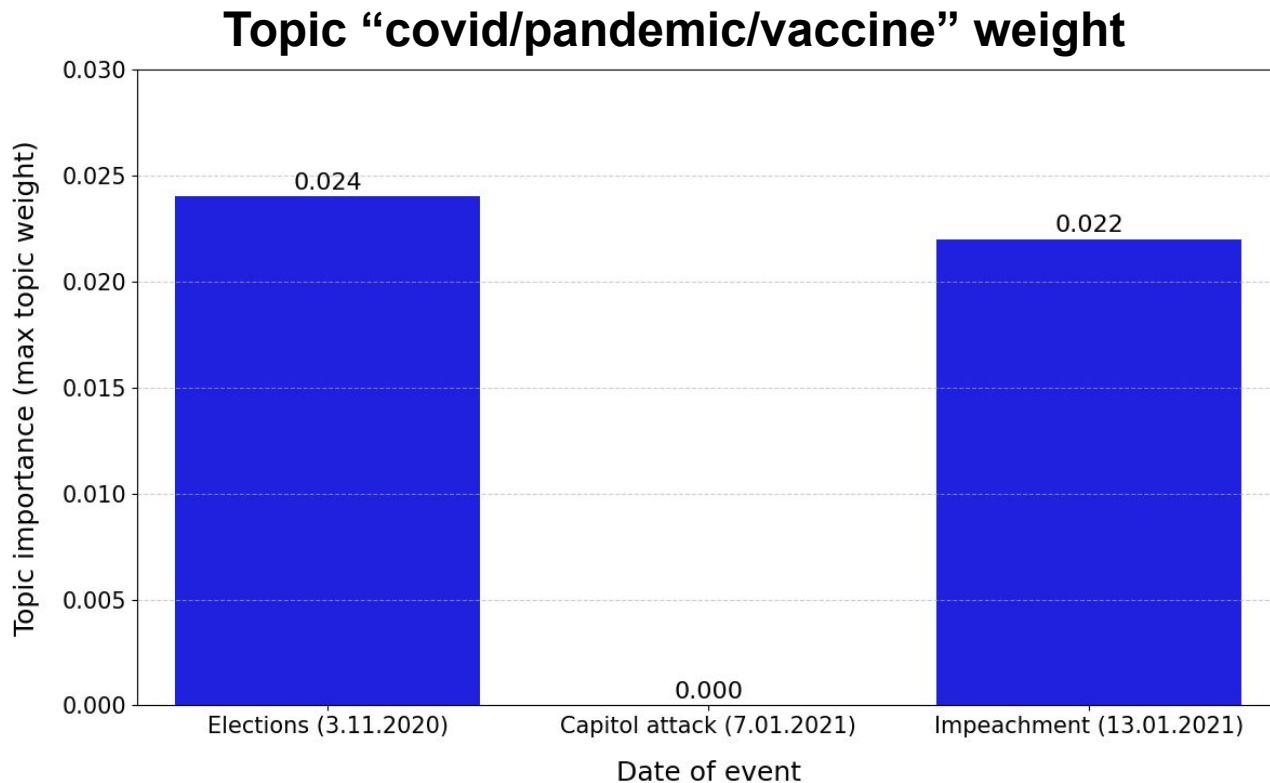
Conservatives used words like "Election Fraud" very often on that day. It means they were trying to justify the reason for the attack, rather than just discussing the violence like other groups

Sentiment distribution by comment length



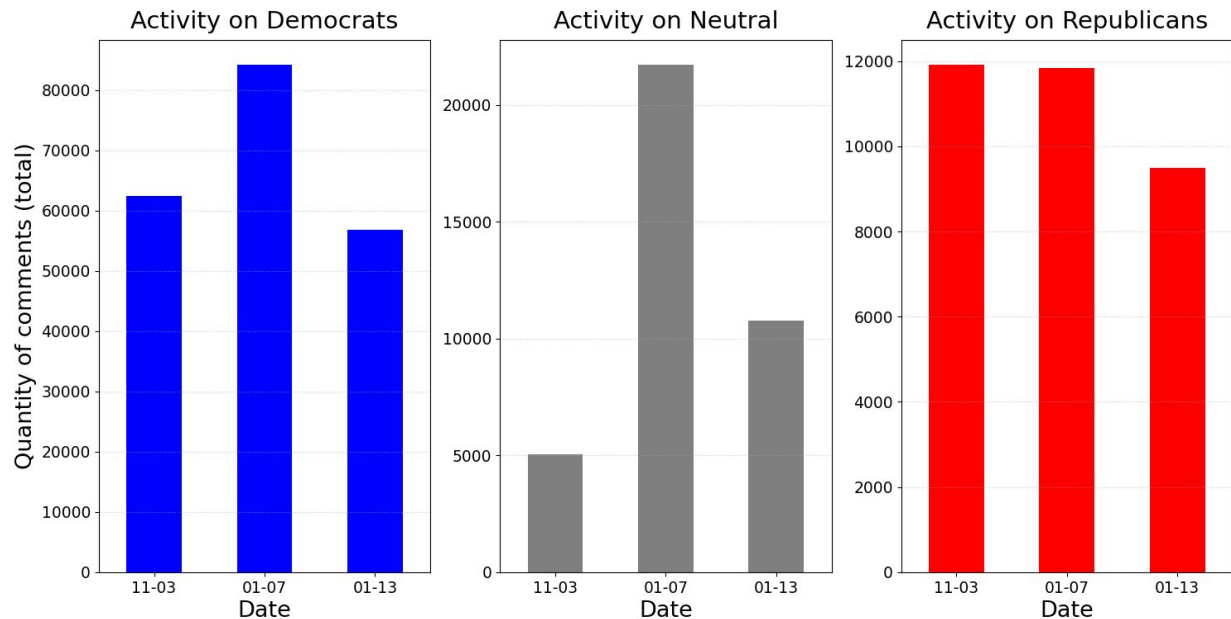
- The highest density of activity (bright yellow/orange) is concentrated in very short comments (under 30 words). This confirms that Reddit users mostly write small statements instead of long texts
- As comments get longer, the sentiment tends to move to the center (neutral). It means that the more time users spend writing, the less emotional the text they produce.

What required deeper investigation



We compared COVID discussions to Election discussion and found that during Capitol attack the topic of COVID completely disappeared

Final insights. Distribution insight



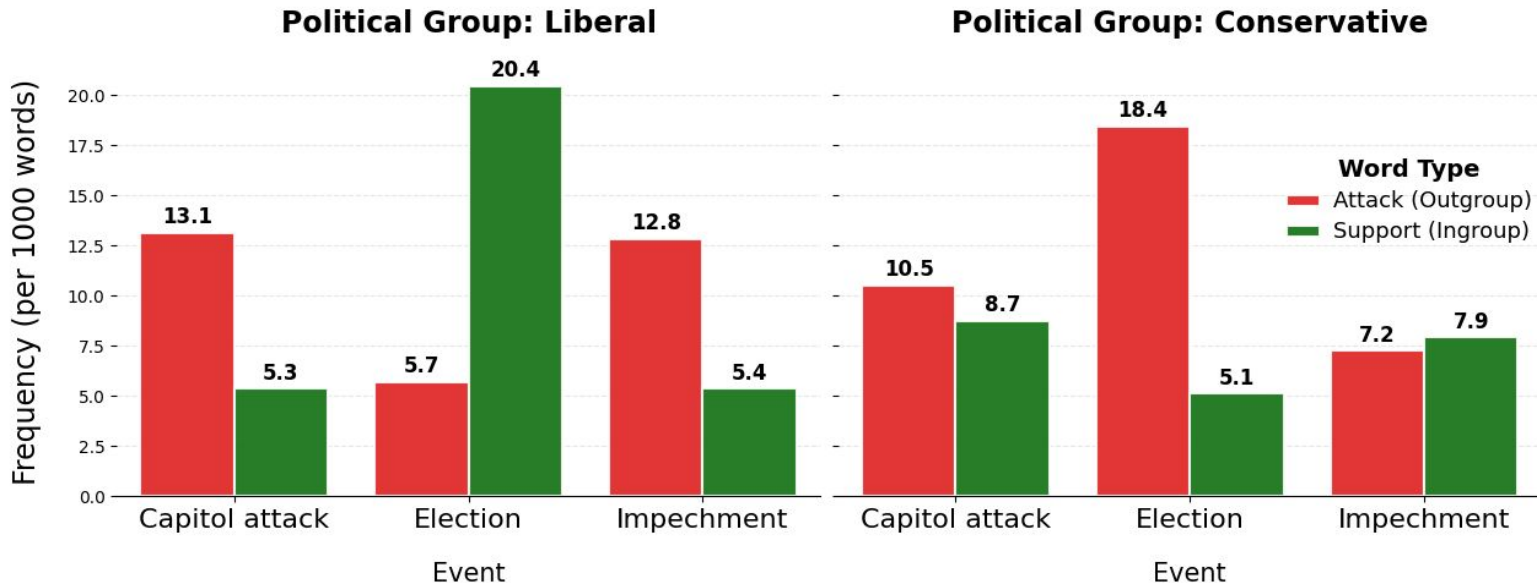
Context of dates:
11-03: Election day, 01-07: Capitol attack, 01-13: Trump's impeachment

Explanation: this plot shows distribution of activity during major events

We can state that neutral users have interest only in extreme events, when left and right wings actively discussed all events

Word type analysis

Explaining: both wings used attack words more than support. Especially during extreme events.

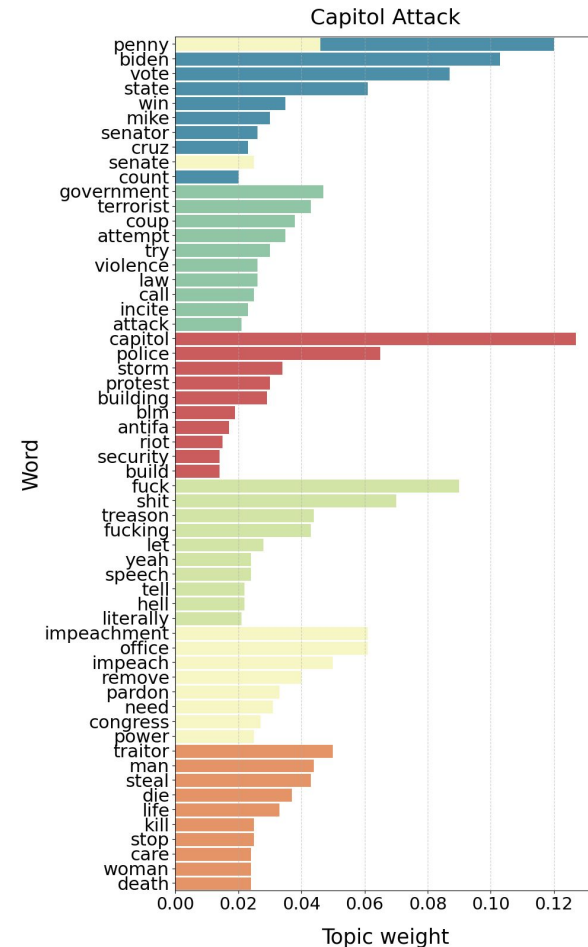
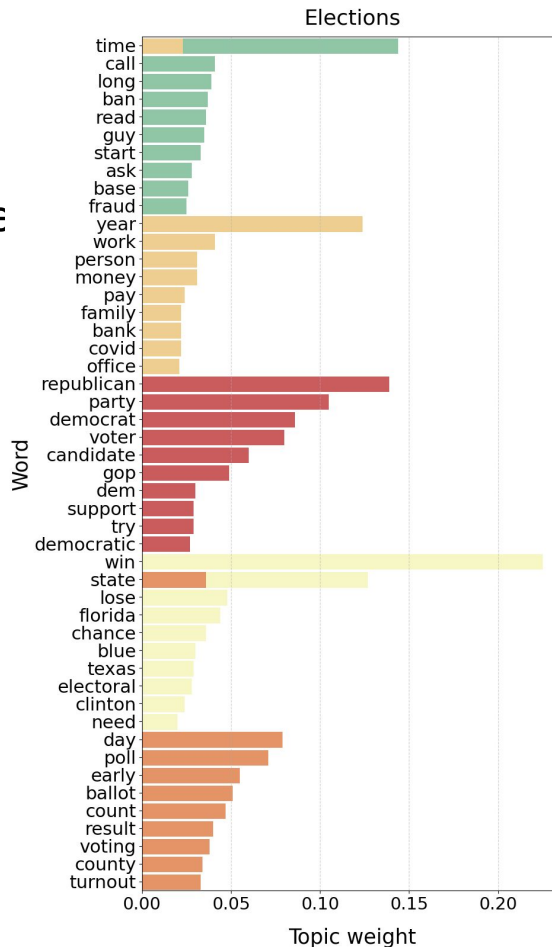


During crises, engagement is powered but by shared enemies. The volume of “attack words” consistently overtakes “support words”. It proves that people on Reddit are mobilized more by hating the enemy than by supporting their own side.

Topic modelling

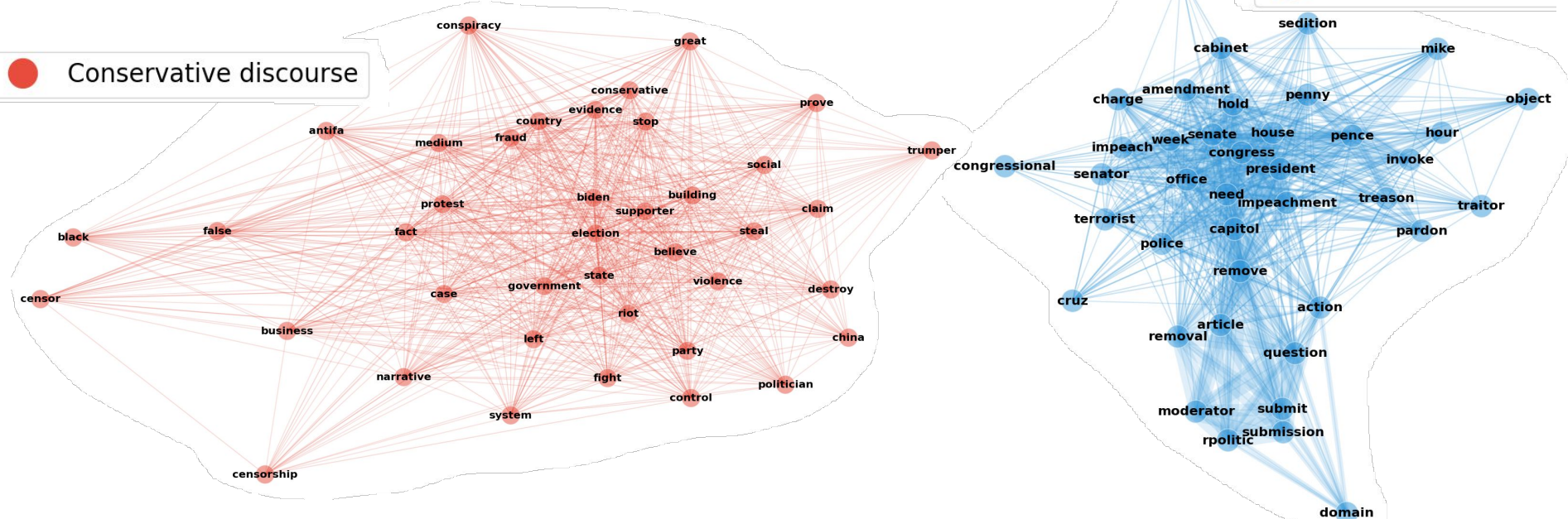
Explanation: The discourse around the elections was simple. Phrases like “win”, “poll” and “ballot” tell us that it was a political process with no surprises. After capitol attack usual lexis changes cardinally to violent.

This proves that during the capitol attack Reddit reacted on event as a crisis, followed with high emotional words. The Reddit community defined this event as a crime against all country rather than a common political protest.



Semantic network analysis

Explanation: This network graph visualizes the distance between the two political groups. It forms two completely separate graphs. Edges indicate the frequency of the two words usage together.



Conservative Discourse: Fewer relations between key terms shows division inside the group. Meaning conservatives are divided on Reddit.

Liberal Discourse: High relations indicating a unified community. Meaning liberals are unified on Reddit.

Future work

Some questions left unanswered:

By analyzing users' interaction, segmenting them and defining patterns:

- Who are the most active users?
- What users had the most influence on others opinion?
- How many bots took part in the comments discussion?
- What is the main characteristic of a bot-made comment?

By analysing material from different sources:

- Can reddit users' behaviour reflect real political situation or is it unique for this community?
- Is reddit the only place that has democrats majority, or it is common for other social networks?

Github repository link

Here you can find all source files, scripts and notebooks with requirements:

[reddit_political_behaviour](#)